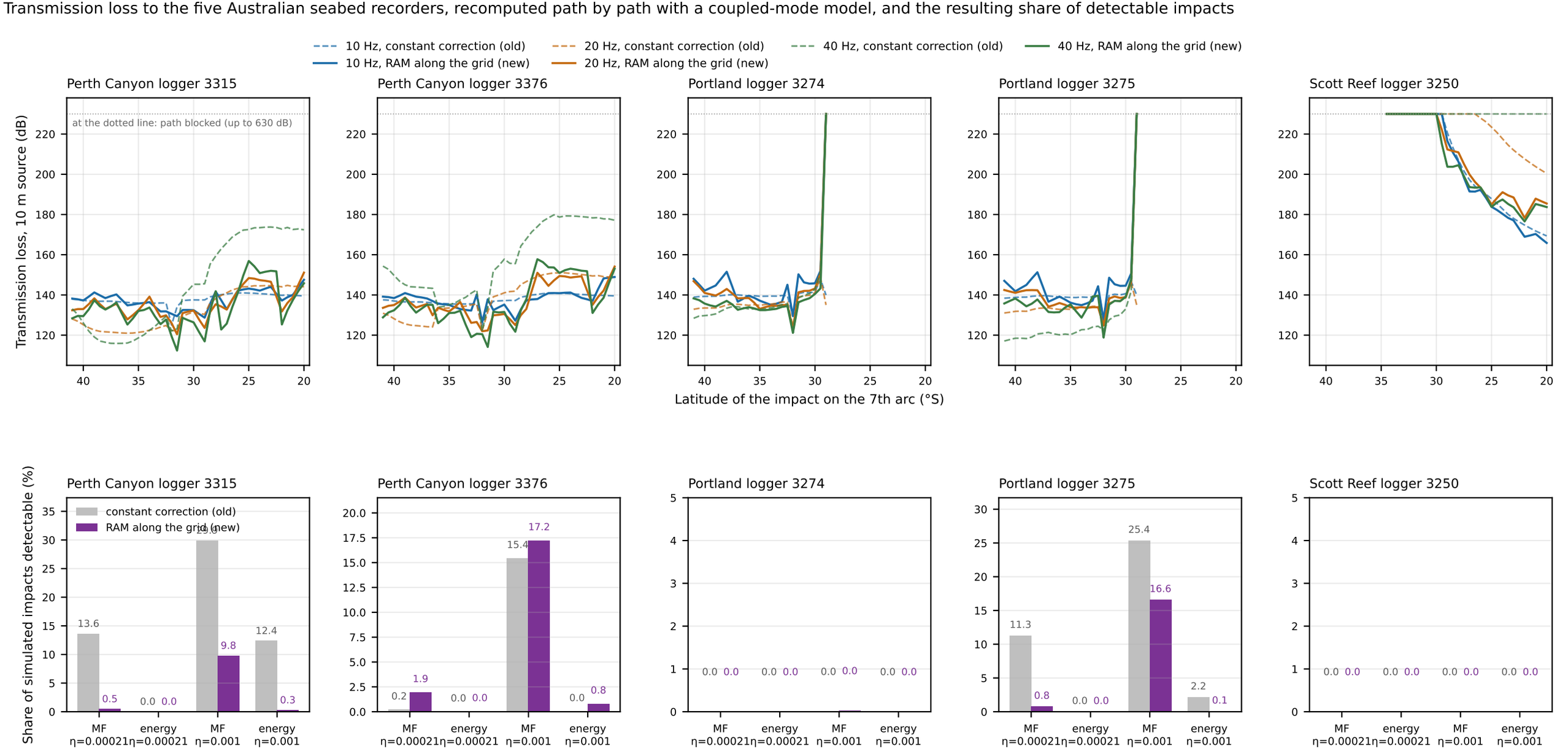

Plain-language note: Top row: how much the sound weakens on its way from an impact at each latitude on the 7th arc to each seabed recorder, in three frequency bands. The dashed lines are the old tables, which applied one fixed correction per recorder; the solid lines are the new tables, computed path by path with a model that handles sloping seabeds. Bottom row: the share of simulated impacts that would be detectable at each recorder (MF = matched-filter detector, 7 dB; energy = conventional energy detector, 15 dB) for two assumed values of the share of impact energy that becomes sound, before (grey) and after (purple) the new tables. Model predictions, not detections; the sound-loss model is not calibrated against a real event at these recorders.

Technical note: source posterior = run J (D0′ 00:19 held out, p = 0.5, 2 M weighted draws, prereg c360db5); old = imos_tl_ramcorr_arc.csv (rule 2 constant Δ from imp500 paths); new = imos_tl_ram_arc.csv = raw adiabatic KRAKEN − Δ_noabs(lat, offset, f), Δ from pyram split-step Padé (dr 2λ, dz λ/20, 9 freqs per third octave, hard bottom cp 1650 / ρ 1.9 / 0.5 dB/λ, no absorption in either model) at 1.0° × offset 0 nodes, 10/20/40 Hz, log-f interpolation clamped (5–8 Hz take the 10 Hz Δ), 10 m source, receiver on its own bathymetry at seabed − 2 m; rows marked × keep the old value. η ∈ {2.1e-4 (F-35A anchor), 1e-3}; thresholds from the weak-signal track. PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated). Module commit cc943df.